DemoShield: Local Noise for Private Sensor Aggregation

I. LOCAL PERTURBATION

This is an original synthetic document for software demonstrations. DemoShield and ToyCounts are fictional. It describes a simplified mechanism, not a proof or deployed privacy guarantee.

DemoShield clips each sensor reading into a fixed interval before adding local Laplace noise. A device sends only its noisy value to the server. The server computes the average of received values and does not receive the original individual readings.

The noise scale is tied to the clipping interval and a privacy parameter. This software example does not specify a numerical privacy parameter or prove a composition bound. It must not be described as a certified privacy system.

DemoShield: Local Noise for Private Sensor Aggregation

II. EVALUATION AND LIMITATIONS

ToyCounts is a fictional set of scalar counts for demonstration. Evaluation compares the aggregate of perturbed counts with the reference aggregate of unperturbed counts. It does not evaluate participant recruitment or missing-reading interpolation.

Increasing noise can reduce the utility of the aggregate. A small number of participants makes individual noise harder to average out. These are qualitative limitations; the document claims no numerical improvement.

The example does not discuss homomorphic encryption, federated optimization or cryptographic security proofs.